

Re: Application for the EngD position, Developing an Interactive Reverse Logistics Map for Circular Construction (CIRCOLOGIC), University of Twente

Dear Professor Hans Voordijk, Professor Berkhout, Professor Marc, and Members of the Selection Committee,

The construction sector's shift from a linear "take, make, use, dispose" model to a circular one depends on getting valuable materials back into use, yet reuse still fails most often for a simple reason: information about a material's availability, quality, location, and future demand is scattered across stakeholders. Reverse logistics is the missing link, and CIRCOLOGIC's plan to build an interactive map, a calculation module, and a scenario engine is exactly the kind of practical decision-support I want to help create. That is why this EngD immediately stood out to me.

My route to this work began with my Master's thesis on Agile Project Management in construction, which taught me that transforming the sector depends as much on coordination and transparent information flow as on technical innovation. That insight pulled me toward circular construction and a question I kept returning to: what happens to materials at end-of-life, and how do we make reliable information about them available to everyone who needs it? It moved my focus from delivering projects to designing the digital tools that make circular decisions possible.

My professional experience maps directly onto this challenge. Building digital twins taught me to organise and integrate data from physical assets into decision-support systems. At Hive CPQ I managed large datasets, gathered requirements from manufacturers and customers, and turned complex rules into configurators that people actually used. Consulting on Agile delivery showed me how to coordinate work across the mixed stakeholder networks circular construction relies on, from contractors and designers to logistics providers, recyclers, municipalities, and clients. Together with my civil and environmental engineering background and GIS skills, this lets me contribute to both the platform's technical development and its real-world adoption.

Most recently, I have taken a deliberate career break to care for my newborns, staying close to the field through self-study in circular construction, reverse logistics, and GIS. This EngD is the ideal way back into professional work: a programme where I can not only refresh the civil, environmental, and digital skills I built earlier, but deepen them and apply them to a genuine industry problem rather than in theory alone.

That is also what draws me to the University of Twente, with its focus on solving real societal challenges alongside industry and an EngD structure that pairs advanced education with a hands-on design project with partners such as Dura Vermeer and TNO. I learn best when I can put new methods to work immediately, which is exactly what this position offers.

I would be honoured to contribute to CIRCOLOGIC and would welcome the opportunity to discuss how my background and motivation fit the project. Thank you for considering my application.

Yours sincerely,

Pooja Awasare